

Sri Chaitanya IIT Academy.,India.

A.P T.S KARNATAKA TAMILNADU MAHARASTRA DELHI RANCHI

A right Choice for the Real Aspirant

ICON Central Office - Madhapur - Hyderabad

Sec:Sr.Super60_NUCLEUS BT **Paper -2(Adv-2024-P2-Model)** **Date: 24-11-2025**

Time: 02.00Pm to 05.00Pm **GTA-06** **Max. Marks: 180**

24-11-2025_Sr.Super60_NUCLEUS-BT_Jee-Adv(2024-P2)_GTA-06_Syllabus,

MATHEMATICS : Total Syllabus
PHYSICS : Total Syllabus
CHEMISTRY : Total Syllabus

Name of the Student: _____ **H.T. NO:** | | | | | | | |

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JEE-ADVANCED-2024-P2-Model

Time:3Hr's **IMPORTANT INSTRUCTIONS** **Max Marks: 180**

MATHEMATICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 1 – 4)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 5 – 7)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 8 – 13)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 14 – 17)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

PHYSICS:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 18 – 21)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 22 – 24)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 25 – 30)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 31 – 34)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

CHEMISTRY:

Section	Question Type	+Ve Marks	- Ve Marks	No.of Qs	Total marks
Sec – I(Q.N : 35 – 38)	Questions with Single Correct Choice	+3	-1	4	12
Sec – II(Q.N : 39 – 41)	Questions with Multiple Correct Choice with partial mark	+4	-2	3	12
Sec – III(Q.N : 42 – 47)	Questions with **NON-NEGATIVE INTEGER** Answer Type	+4	0	6	24
Sec – IV(Q.N : 48 – 51)	Paragraph with NUMERICAL VALUE	3	0	4	12
Total				**17**	**60**

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MATHEMATICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

This section contains **FOUR (04)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

==========

1. Area bounded between the curves $y=\sqrt{4-x^2}$ and $y^2=3|x|$ is/are

A) $\frac{\pi-1}{\sqrt{3}}$ **B)** $\frac{2\pi-1}{3\sqrt{3}}$ **C)** $\frac{2\pi-\sqrt{3}}{3}$ **D)** $\frac{2\pi-\sqrt{3}}{3\sqrt{3}}$

2. Mr. A makes a bet with Mr. B that in a single throw with two dice he will throw a total of seven before B throws four. Each of them has a pair of dice and they throw simultaneously until one of them wins equal throws being disregarded. Probability that B wins, is

A) $\frac{1}{3}$ **B)** $\frac{4}{11}$ **C)** $\frac{5}{16}$ **D)** $\frac{6}{17}$

3. If $\sin^{-1}\left(\sin\frac{33\pi}{7}\right)+\cos^{-1}\left(\cos\frac{46\pi}{7}\right)+\tan^{-1}\left(-\tan\frac{13\pi}{8}\right)+\cot^{-1}\left(\cot\left(-\frac{19\pi}{8}\right)\right)=\frac{a\pi}{b}$

Where a and b are in their lowest form, then (a + b) equal to

A) 17 **B)** 20 **C)** 23 **D)** 22

4. Let $z_r\,(1\le r\le 4)$ be complex numbers such that $|z_r|=\sqrt{r+1}$ and

$$\left|30z_1+20z_2+15z_3+12z_4\right|=k\left|z_1z_2z_3+z_2z_3z_4+z_3z_4z_1+z_4z_1z_2\right|$$

Then the value of k equals

A) $|z_1z_2z_3|$ **B)** $|z_2z_3z_4|$ **C)** $|z_4z_1z_2|$ **D)** $|z_1z_3z_4|$

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SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

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5. In a gambling between Mr. A and Mr. B a machine continues tossing a fair coin until the two consecutive throws either HT or TT are obtained for the first time. If it is HT, Mr, A wins and if it is TT, Mr, B wins. Which of the following is (are) true?

A) probability of winning Mr.A is $\frac{3}{4}$

B) Probability of Mr.B winning is $\frac{1}{4}$

C) Given first toss is head probability of Mr. A winnings is 1

D) Given first toss is tail, probability of Mr.A winning is $\frac{1}{2}$

6. The projection of line $3x-y+2z-1=0=x+2y-z-2$ on the plane $3x+2y+z=0$ is

A) $\frac{x+1}{11}=\frac{y-1}{-9}=\frac{z-1}{-15}$ **B)** $3x-8y+7z+4=0=3x+2y+z$

C) $\frac{x+12}{11}=\frac{y+8}{-9}=\frac{z+14}{15}$ **D)** $\frac{x+12}{11}=\frac{y+8}{-9}=\frac{z+14}{-15}$

7. Let $f(x)=\begin{cases}2x-4 & x\le 2\\ -x^2+\frac{k^3(k-1)^2}{k^2-k-2}+4, & x>2\end{cases}$. $f(x)$ attains local maximum at x = 2 if k lies in

A) (0, 1) **B)** $(3,\infty)$ **C)** $(-\infty,-1)$ **D)** (1, 2)

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SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

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8. In triangle ABC, with AB < AC, let H be the orthocenter and O be the circumcentre. Given that mid point of OH lies on BC, BC = 1 and the perimeter of ΔABC is 6 units then the area of Δle ABC is $\frac{a}{b}$ sq. units (GCD(a, b) = 1) then a + b is equal to …….

9. The number of ordered pair(s) (a, b) for which the function $f(x)=\operatorname{sgn}\left(\left(x^2-ax+1\right)\left(bx^2-2bx+1\right)\right)$ is discontinuous at exactly one point 1 (Where a, b are integers and sgn(x) denotes signum function of x) is/are ……..

10. $\sum_{K=0}^{\infty}\frac{(-1)^K}{(K+1)(2K+1)3^K}=\frac{\pi}{\sqrt{a}}+3\log_e\left(\frac{a}{b}\right)$,(where *a* is prime number and *b* is real number). The value of a^2+b^2 is equal to ……

11. Let $f(x)=\lim_{n\to\infty}\frac{\log_e\left(e^n+x^n\right)}{n}$, $x\geq 0$, then the area bounded by y = f(x) and x = 0, $x=e^2$ and $x-$axis is e^m+e^n sq. units. The value of m + n is equal to ……

12. Let $P=\{z\in C:|z+2-3i|\leq 1\}$ $Q=\{z\in C:z(1+i)+\bar{z}(1-i)\leq -8\}$. Let in $P\cap Q, |z-3+2i|$ be maximum and minimum at z_1 and z_2 respectively. If $|z_1|^2+2|z_1|^2=\alpha+\beta\sqrt{2}$, where α, β are integers, then $\frac{\alpha+\beta}{4}$ equals ……..

13. Let {x} denote the fractional part of 'x' and $f(x)=\frac{\cos^{-1}\left(1-\{x\}^2\right)\sin^{-1}\left(1-\{x\}\right)}{\{x\}-\{x\}^3}$, $x\neq 0$. If L and R respectively denotes the left hand limit and the right hand limit of f(x) at x = 0, then $\frac{16}{\pi^2}\left(L^2+R^2\right)$ is equal to …….

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SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

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PARAGRAPH-I

Let $\left(1+x+x^2\right)^{20} = a_0 + a_1x + a_2x^2 + \ldots\ldots\ldots + a_{40}x^{40}$. Further $S = \sum_{r=0}^{40} a_r$. Two coefficients are chosen from the coefficients $a_0, a_1, \ldots\ldots\ldots, a_{40}$ and the probability that they are equal is p (considering no three coefficients are equal). Units digit of S is equal to b and $a = \frac{1-p}{10p}$.

On the basis of above information, answer the following

14. If $\left(\sqrt{a-b}+b\right)^6 = I + F$, where $0 \le F < 1$ and $I \in N$, then the value of I is

15. The coefficient a_3 is equal

PARAGRAPH-II

Consider the matrices $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}; B = \begin{bmatrix} \frac{1}{2} & 3 \\ 0 & 1 \end{bmatrix}$

Let P be an orthogonal matrix and
$Q = PAP^T, R_K = P^TQ^KP, S = PBP^T$ and $T_K = P^TS^KP$. Where $K \in N$.

16. The value of $\sum_{K=1}^{5} a_K + \sum_{K=1}^{3} b_K$ is ……. Where a_K represents the element of first row and first column in matrix R_K and where b_K represents the element of second row & second column in matrix R_K

17. The value of $\sum_{K=1}^{\infty} X_K + \sum_{K=1}^{10} Y_K$ is $\frac{m}{n}$, where m, n are co-primes .Then the value of $(56n - m)$ is ……

(X_K represents the element of first row & first column in matrix T_K and where Y_K represents the element of second row & second column in matrix T_K)

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PHYSICS **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

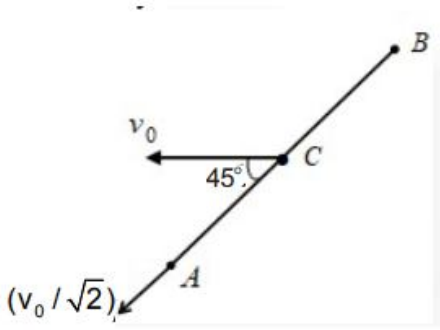
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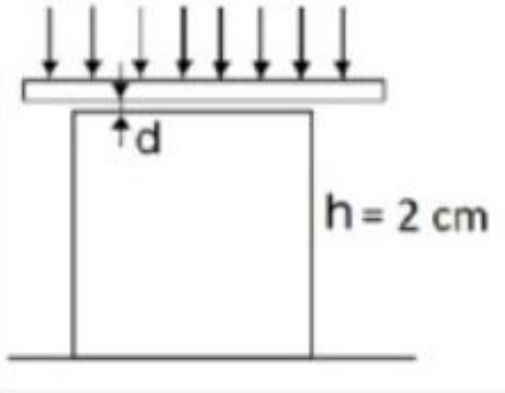
18. A rigid rod AB moves on horizontal plane such that at the instant shown velocity of centre C of rod is v_0 at an angle of 45^0 with the rod. While velocity of end A is $v_0\sqrt{2}$ and has direction along the rod. Velocity of end B of the rod at the given instant is



A) $v_0/\sqrt{2}$ **B)** $2v_0$ **C)** $\sqrt{\frac{5}{2}}v_0$ **D)** $\sqrt{2}v_0$

19. A glass plate is placed above a glass cube of 2 cm edges in such a way that there remains a thin air layer of thickness 'd' between them (see figure). Electromagnetic radiation of wavelength between 400 nm and 1150 nm (for which the plate is penetrable) is incident perpendicular to the plate from above, is reflected from both air surfaces (bottom of the glass plate and top of the cube) and interferes. In this range, only two wavelengths give maximum reinforcements, one of them is 400 nm. λ is the second wavelength. T is change in temp. necessary to warm up the cube so as it would touch the plate. The coefficient of linear thermal expansion of material of cube is $\alpha = 8\times10^{-6}/\,^0C$, the refractive index of the air $\mu = 1$. The distance of the bottom of the cube from the plate do not change during warming up. Then

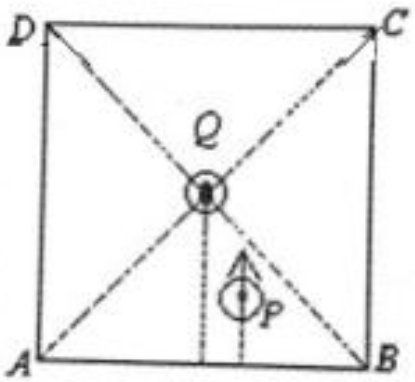
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The value of λ (in nm) is

A) $\frac{200}{3}$ **B)** $\frac{4000}{3}$ **C)** 1000 **D)** $\frac{2000}{3}$

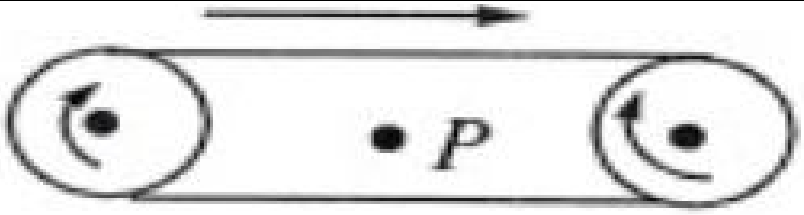
20. ABCD is a smooth horizontal square of side L. A cylindrical smooth coin-Q of radius 'r' is at rest at the centre of the square. Another cylindrical smooth coin-P of radius 2r and moving parallel to AD and strikes the coin-Q. The coin-Q after the collision moves towards D. The distance between the line of motion of the centre of coin-P and the line AD is



A) $\frac{L}{2}+\frac{3r}{\sqrt{2}}$ **B)** $\frac{L}{2}+\sqrt{2}r$ **C)** $\frac{L}{2}+\sqrt{2}r$ **D)** $\frac{L}{2}+\frac{9r}{5}$

21. A wide, very long, insulating belt has a uniform positive charge per unit area σ on its outer surface. Rollers at each end starts rotating from rest with variable angular acceleration, $\alpha = at^2 + bt + c$. Where a, b, c are constants and 't' is time elapsed from the starting of the motion. There is no slipping between the belt and the rollers. The radii of the rollers is r and the length of the belt is L. r << L. The magnitudes of magnetic field at P (a point in between the belts) and Q (a point outside and close to the belt) as shown into the diagram at a time 't' respectively are neglect the edge effect ie., the length of the belt which is contact with pulley.

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A) (0, 0)

B) $\frac{\mu_0 r\sigma}{6}\left(2at^3+3bt^2+6ct\right),0$

C) $0, \frac{\mu_0 r\sigma}{12}\left(2at^3+3bt^2+ct\right)$

D) $\frac{\mu_0 r\sigma}{3}\left(2at^3+3bt^2+ct\right), \frac{\mu_0 r\sigma}{9}\left(2at^3+3bt^2+ct\right)$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

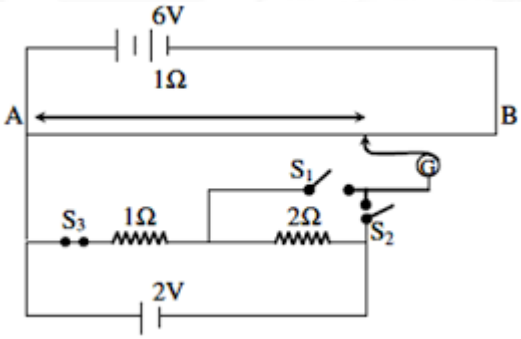
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22. Figure shows a potentiometer circuit. The length of the potentiometer wire AB is 120 cm and its resistance is 2Ω . The internal resistance of 2V cell is negligible and the internal resistance of 6V cell is 1Ω . Initially the switches S_1 and S_2 are open and S_3 is closed



A) When only S_2 is open balanced length $l = 20$ cm

B) When only S_1 is open balanced length $l = 60$ cm

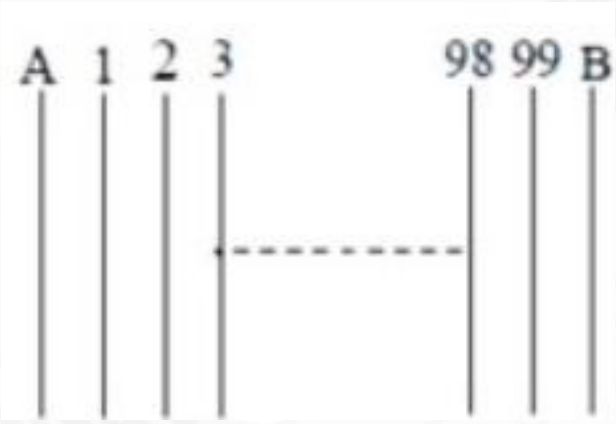
C) When all three switches are closed balanced length 60 cm

D) When S_3 is open, S_2 is also open but S_1 is closed balance length is 60 cm

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23. Two perfectly black parallel plates A and B of equal area facing each other are maintained at constant temperature T_A and $T_B (T_A > T_B)$. And 99 perfectly black and conducting identical screens of same area as plates are introduced between A and B which are parallel to each other having equal space between them. Once the thermal equilibrium is reached. Choose the CORRECT statements:

(take $\sigma\left(T_A^4 - T_B^4\right) = 5000 \frac{W}{m^2}$: ignore convection)



A) Heat flux between screen 10 and 2 is greater than heat flux between screen 98 and 99

B) Heat flux between plate A and screen 1 is 5000 $\frac{W}{m^2}$

C) Heat flux between screen 99 and plate B is 50 $\frac{W}{m^2}$

D) Effective heat flux between plates A and B reduces due to introduction of screens between them

24. There is a point source of light at a depth 8m below the surface of water. An observer in air sees the image of the source. Line of sight of observer makes an angle 53^0 with the normal to surface. (use: $\sqrt{7} = 2.65$ and $\mu =$ refractive index of water = 4/3)

A) Apparent depth of image from surface is $\frac{71}{32}$ m for the observer

B) Horizontal distane between object and its image is 3.38m

C) Radius of circle of illuminance is 9.05m

D) Percentage of power coming out of the with in the circle of illuminance is 16.875% (assume no absorption of light by water and 100% refraction at surface)

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SECTION-III
(NON-NEGATIVE INTEGER)

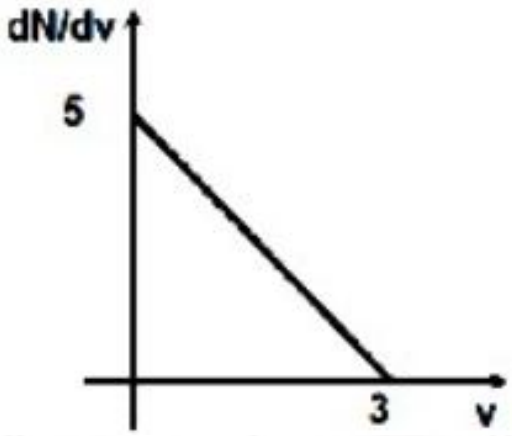
- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases..*

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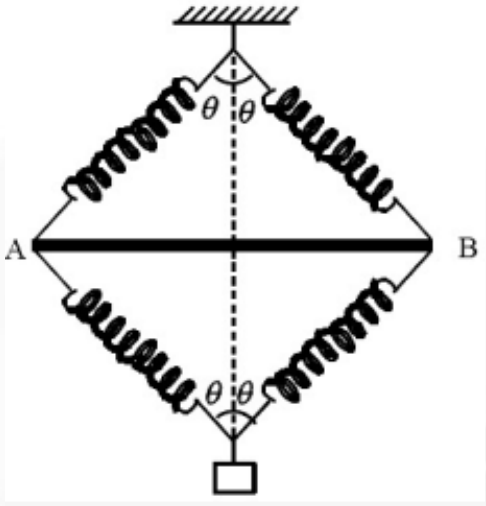
25. A soap bubble of radius 'r', surface tension T exists in equilibrium in vacuum. A Mono-atomic gas is trapped inside the soap bubble. Charge Q is given to bubble such that bubble expands from radius 'r' to radius 2r under adiabatic condition. If surface charge density is found to be $\sqrt{\frac{(4x-1)\varepsilon_0 T}{xr}}$. Find the value of 'x'

26. Consider a gas kept in a closed container. 'N' is the no of molecules present in the container and 'v' is the speed of molecules. Molecular speed distribution graph is shown in the figure with suitable S.I. units of $\frac{dN}{dv}$ and *v*. The v_{rms}^2 of molecules in S.I units is $\frac{x}{2}$, then x = ……



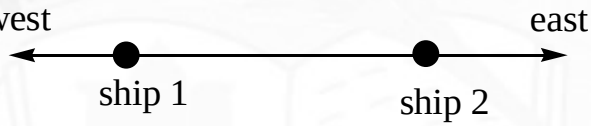
27. Four highly stiff identical springs each of spring constant K are connected as shown in figure. Distance between joints 'A' and 'B' is kept constant by fixing a rigid, light horizontal rod. A block of mass *m* connected at lower end is in equilibrium. Now block is displaced slightly in vertical direction and released. Time period of oscillation of block $T = 2\pi\sqrt{\frac{mx}{2K}}$. Fill the value of *x* in OMR sheet, take $\theta = \frac{\pi}{4}$

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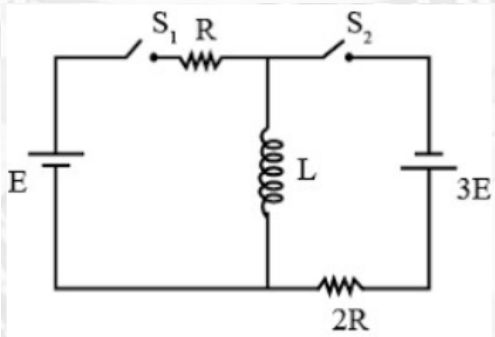


28. Intensity observed in an interference pattern is $I = I_0 \sin^2\theta$. At $\theta = 30^0$, intensity $I = 5 \pm 0.0020$ W / m^2. If $I_0 = 20$ W / m^2 and percentage error in angle θ is $\frac{n}{\pi}\sqrt{3} \times 10^{-2}\%$, then the value of 'n' is

29. Two ships are 10 km apart on the west-east line at some instant, the first one streaming to the east at 30 km/h and the second one to north at 40 km/h. what is the distance of closest approach between them in km?



30. In the circuit shown, switch S_2 is open and S_1 is closed since long. Take E = 20V, L = 0.5 H and $R = 10\Omega$. The magnitude of rate of change Energy of stored in the magnetic field inside the inductor, immediately after S_2 is closed is 'N' J/s then $\frac{N}{10}$ is ……



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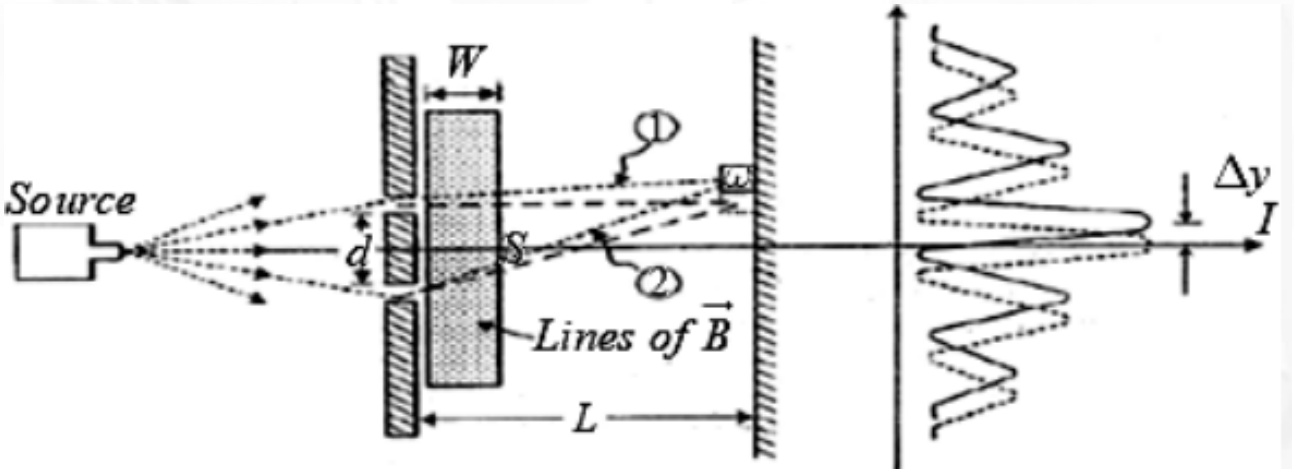
SECTION-IV
(PARAGRAPH QUESTIONS)

- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

==========

PARAGRAPH-I

Bohm Aharonov effect : Consider a YDSE experiment done with electrons. The debroglie hypothesis says that electrons behave like waves and so in presence of double slit, they show interference pattern as shown by dotted line. But if we introduce a thin strip (w < < L) of magnetic field just in front of slits, the electrons trajectory would bend by a small angle and so the central maxima shifts up by a distance Δy . This shift can be easily shown to depend on magnetic field B, charge q, width *w*, the momentum of electron *p*, slit distance d and L. The same result can be derived using principles of quantum mechanics. In quantum mechanics we cannot talk about definite path. So we cannot talk about bending of electron trajectory. There it is found that on passing through the region of magnetic field, the electrons waves undergo a phase change. So the electrons wave coming from upper slit and that from lower slit have a phase difference. Interestingly the path difference resulting from this phase difference predicts the same shift of central maxima Δy as does the classical theory using magnetism



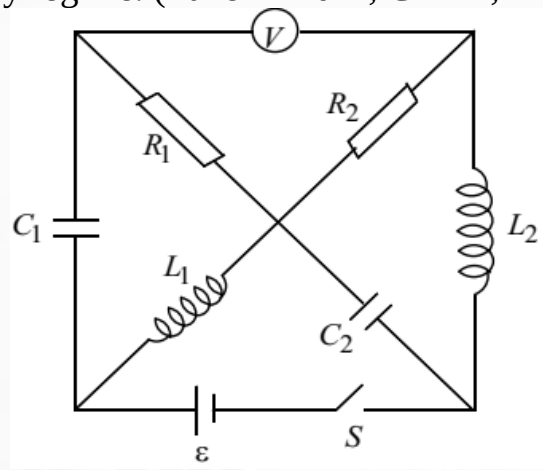
31. The value of Δy according to classical physics, if electron momentum is 10^{-25} kgm/s, B = 1T, $w = 10^{-8}$m, L = 1m, d = 500 nm is $4\times(\text{x mm})$. Find 'x'

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32. In a typical set up, electrons a momentum of 10^{-25} kgm / s, $B = 1T$, $w = 10^{-8}$m, $L = 1m$ at what minimum value of d (in nm) will central point be a minimum? $\left(\text{Take } h = 6.4 \times 10^{-34} \text{ J. sec.}\right)$

PARAGRAPH-II

For the circuit shown in figure, $R_1 = 3R$, $R_2 = R$, $C_1 = C_2 = C$ and $L_1 = L_2 = L$. The electromotive force of the battery is ε. Initially the switch is closed and the system is operating in a stationary regime. (Take R=10Ω, C = 1F, L = 1H, $\varepsilon = 20V$)



33. Reading of the ideal voltmeter (in volt) immediately after the switch is open is equal to ……

34. Total amount of heat dissipated (in Joules) after opening the switch, until a new equilibrium state is achieved, is equal to ……

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CHEMISTRY **Max Marks: 60**

SECTION–I
(SINGLE CORRECT ANSWER TYPE)

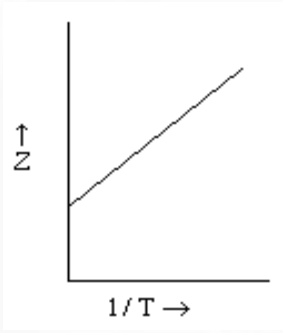
This section contains **FOUR (04)** questions.

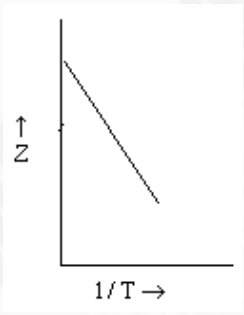
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.
- For each question, choose the option corresponding to the correct answer.
- Answer to each question will be evaluated according to the following marking scheme:

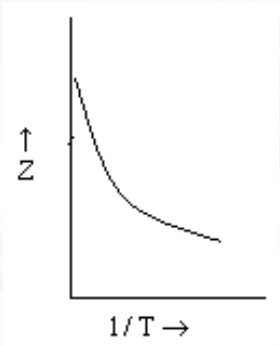
Full Marks : +3 If **ONLY** the correct option is chosen;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : −1 In all other cases

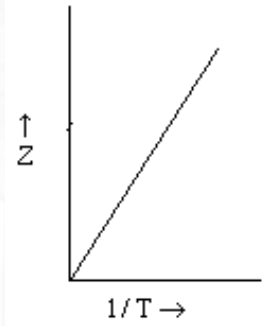
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35. Which of the following graphs is correct for a sample of 1 mole of a real gas at constant high pressure?

A) 

B) 

C) 

D) 

36. Which of the following statement is wrong regarding copper sulphate?

A) It reacts with KI to give I_2.

B) It reacts with KCl to give Cl_2.

C) It produces Fehling's solution reacts with glucose gives red colour ppt.

D) It gives CuO on strong heating in air.

37. Spin magnetic moment of X^{3+} ion of 4d series is $2\sqrt{6}$ BM. What is the atomic number of X?

A) 42 **B)** 44 **C)** 43 **D)** 46

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38. $CO_2 + NH_3 + H_2O \rightarrow X;\ X + NaCl \rightarrow Y + NH_4Cl;\ 2Y \xrightarrow{\Delta} Z + H_2O + CO_2$

Compound Z is :

A) Na_2CO_3 **B)** $(NH_4)_2CO_3$ **C)** $(NH_4)HCO_3$ **D)** $NaHCO_3$

SECTION – II
(ONE OR MORE CORRECT ANSWER TYPE)

This section contains **THERE (03)** questions.

- Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).
- For each question, choose the option(s) corresponding to (all) the correct answer(s).
- Answer to each question will be evaluated according to the following marking scheme :

Full Marks :+4 **ONLY** if (all) the correct option(s) is(are) chosen;
Partial Marks: +3 If all the four options are correct but **ONLY** three options are chosen;
Partial Marks: +2 If three or more options are correct but **ONLY** two options are chosen, both of which are correct ;
Partial Marks: +1 If two or more options are correct but **ONLY** two options are chosen, and it is a correct option ;
Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered);
Negative Marks : –2 In all other cases.
For example, in a question, if (A), (B) and (D) are the ONLY three options corresponding to correct answers, then choosing ONLY (A), (B) and (D) will get +4 marks; choosing ONLY (A) and (B) will get +2 marks; choosing ONLY (A) and (D) will get +2 marks; choosing ONLY (B) and (D) will get +2 marks; choosing ONLY (A) will get +1 mark; choosing ONLY (B) will get +1 mark; choosing ONLY (D) will get +1 mark; choosing no option (i.e. the question is unanswered) will get 0 marks; and choosing any other combination of options will get -2 marks.

==========

39. One litre of acidified $KMnO_4$ solution containing 15.8g $KMnO_4$ $(mol.wt = 158)$ is decolorized by passing sufficient amount of SO_2. If SO_2 is produced by FeS_2, then which of the following options is / are correct? $(Fe = 56, S = 32)$

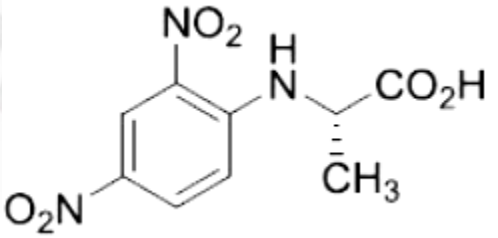
A) Mole of FeS_2 consumed is 0.125

B) SO_2 produced from FeS_2 occupies 5.6 lit at NTP

C) Equivalent weight of $KMnO_4$ in the reaction is 31.6

D) Equivalent weight of SO_2 in the reaction is 64

40. Tetrapeptide(s) that gives (give) the following product on reaction with Sanger's reagent followed by hydrolysis is (are)



A) Ala-Gly-Leu-Phe **B)** Asp-Phe-Leu-Pro

C) Asp-Gly-Tyr-Phe **D)** Ala-Phe-Tyr-Pro

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41. Which of the following reactions liberate nitrogen gas?

A) $NO_2^-(aq) + NH_4^+(s) \xrightarrow{\text{boil}} .....$

B) $CS(NH_2)_2(s) + HNO_2(l) \longrightarrow .....$

C) $NH_3(g) + NaOCl(aq) \xrightarrow{\text{gelatine (or) glue}} .....$

D) $NO_2^-(aq) + Zn(s) + OH^-(aq) \longrightarrow .....$

SECTION-III
(NON-NEGATIVE INTEGER)

- This section contains **SIX (06)** questions.
- The answer to each question is a **NON-NEGATIVE INTEGER**..
- For each question, enter the correct integer corresponding to the answer using the mouse and the on-screen virtual numeric keypad in the place designated to enter the answer.
- Answer to each question will be evaluated according to the following marking scheme:

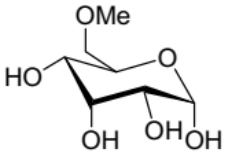
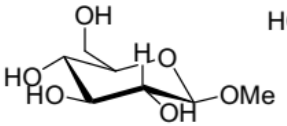
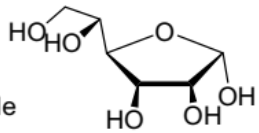
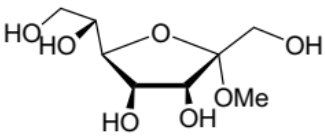
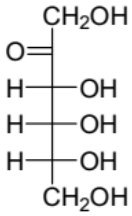
Full Marks: +4If ONLY the correct integer is entered; *Zero Marks : 0 In all other cases.*

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42. A complex compound is represented as $CoCl_3.xNH_3$. Its 0.1M solution in water shows depression in freezing point equal to 0.558K. Assuring 100% ionization of complex and coordination number of Co to be six, calculate the value of x? K_f for H_2O is 1.86 K Kg/mol.

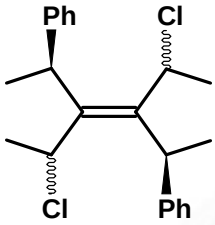
43. Half - life period of the decomposition of a compound is 20 minutes. If the initial concentration is doubled, the half - life period is reduced to 10 mins. What is the order of the reaction?

44. The number of reducing sugars among the following is

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45. The value of [X] in the following reaction is-



$\xrightarrow[(S_N2)]{\text{PhMgBr (excess)}}$ **[X] number of products**

: **Bond configuration is NOT fixed**

: **Bond configuration is fixed.**

46. Find out total number of electrons of Ir^{2+} (atomic number 77) ion which are present in non-axial d-orbitals in the complex compound : $[Ir(dmg)_2]$ is

47. Maximum number of N – Co – O bond angles (90^0) in cis $[Co(gly)_3]$

SECTION-IV
(PARAGRAPH QUESTIONS)

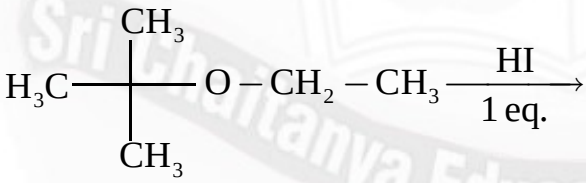
- This section contains **TWO (02)** questions stems.
- There are **TWO (02)** questions corresponding to each question stem.
- The answer to each question is a **NUMERICAL VALUE.**
- For each question, enter the correct numerical value of the answer using the mouse and the onscreen virtual numeric keypad in the place designated to enter the answer.
- If the numerical value has more than two decimal places, **truncate/round-off** the value to **TWO** decimal places.
- Answer to each question will be evaluated according to the following marking scheme:
- ***Full Marks: +3*** If ONLY the correct numerical value is entered at the designated place;
- **Zero Marks:0** in all other cases

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PARAGRAPH-I

A) $CH_3-CH_2-OH \xrightarrow[413\ K]{conc.\ H_2SO_4}$

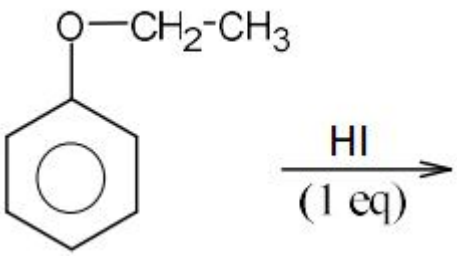
B)



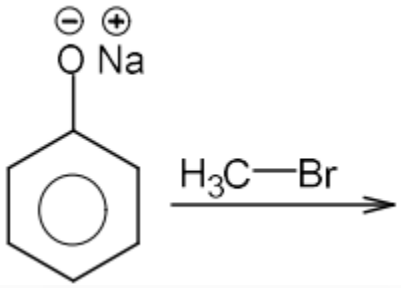
C) $H_3C-CH_2-\underset{\displaystyle CH_3}{\underset{|}{CH}}-CH_2-O-CH_2-CH_3 \xrightarrow{HI(1\ eq)}$

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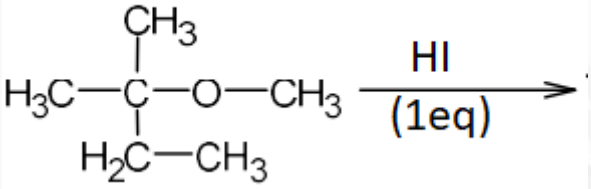
D)



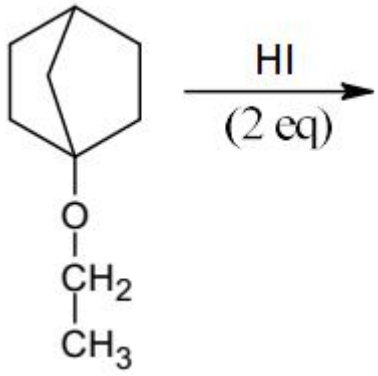
E)



F)



G)



X = Number of reaction(s) in which S_N2 is involved to form major product.

Y = Number of reaction(s) in which S_N1 is involved to form major product.

48. Find the value of x …….

49. Find the value of y ……

PARAGRAPH-II

Strength of the sample of H_2O_2 is generally expressed in terms of volume strength. It means the volume of oxygen liberated at N.T.P by heating one volume of H_2O_2. The concentration of H_2O_2 in a solution can also be expressed as percentage of H_2O_2 in solution. Normality of this solution can be calculated if the equivalent mass of H_2O_2 is known

50. 25 ml of H_2O_2 solution was added to the excess of acidified KI solution. The iodine so liberated required 40 ml of 0.1 N $Na_2S_2O_3$ solution. What is normality of H_2O_2 solution?

51. Percentage strength of the above H_2O_2 solution is

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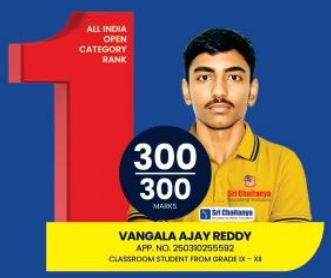
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JEE MAIN 2025

31 STUDENTS BELOW 100 AIR



VANGALA AJAY REDDY
APP. NO. 250310255592
CLASSROOM STUDENT FROM GRADE IX - XII



DEVDUTTA MAJHI
APP. NO. 250310016185
DLP/AITS STUDENT



TOSHNIWAL SHIVEN
APP. NO. 250310391420
DLP/AITS STUDENT



SAKSHAM JINDAL
APP. NO. 250310236696
DLP/AITS STUDENT

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 10 ALL INDIA CATEGORY RANKS COUNT	BELOW 100 ALL INDIA CATEGORY RANKS COUNT	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT
31	95	10	98	579

TOTAL QUALIFIED RANKS FOR JEE ADVANCED-2025 22,094

*DLP/AITS

JEE 2025 STARS SHINE BRIGHT

Sri Chaitanya Tops JEE ADVANCED

ALL INDIA OPEN CATEGORY RANKS



RUTVIK SAI
H.T.No. 256055278 (OBC-NCL)



MAJID MUJAHID HUSAIN
H.T.No. 251134112*



UJJWAL KESARI
H.T.No. 252016104*



AKSHAT KUMAR CHAURASIA
H.T.No. 254065055*

BELOW 100 ALL INDIA OPEN CATEGORY RANKS	BELOW 500 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA OPEN CATEGORY RANKS	BELOW 1000 ALL INDIA CATEGORY RANKS COUNT	NUMBER OF QUALIFIED RANKS
29	113	205	745	4,212

*DLP/AITS

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